

Yumi Heo

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Data Scientist

I build efficient, explainable models tailored to clients' requirements and successfully deploy them into production

SKILLS

Languages & Tools: Python, SQL (PostgreSQL), Git, Docker, FastAPI, Jira, Confluence

ML & CV: PyTorch, TensorFlow/Keras, OpenCV, Torchvision, Pillow, ONNX, Scikit-learn, Pandas, NumPy

LLM & Agentic AI: HuggingFace, LangChain, LangGraph, MCP

Visualisation: Matplotlib, Seaborn, Plotly, Tableau, Folium

Cloud & Big Data: AWS (RDS, EC2, S3), GCP (BigQuery, Compute Engine, Cloud Run, Cloud Storage), PySpark

PROJECTS

- **NoteGuard:** Developed a clinical text de-identification pipeline using rule-based detection, Presidio NER, and patient-consistent pseudonymisation to support privacy-preserving use of free-text clinical data, built at a hackathon
- **Automated-EMA-NICE-Data-Scraping:** Built an automated Python pipeline using GPT-4o mini to extract and summarise unstructured data from the European Medicines Agency, significantly improving data collection efficiency
- **Inventory Placement Optimiser:** Developed a prototype for an inventory placement optimiser using demand forecasting, multinomial logistic regression and linear programming to minimise fulfilment costs under warehouse capacity constraints

WORK EXPERIENCE

Data Scientist, AITIS, London, UK

Jan 2025 – Present

AITIS delivers AI-powered clinical technology for healthcare and consumer smart wearables for health monitoring. In my role, I develop and optimise classification and forecasting models by collaborating with doctors and dermatologists.

- Reduced clinician wound image annotation workload by 90% by developing a sample prioritisation tool that highlighted the hardest-to-classify images for human review
- Developed skin cancer classification models achieving over 90% sensitivity and specificity by addressing class imbalance using CutMix and MixUp augmentation
- Deployed models to production through FastAPI and Docker in collaboration with software engineers
- Mentored summer interns to support their onboarding and personal development

Data Science Intern, Office of Health Economics, London, UK

Jul 2024 – Oct 2024

The Office of Health Economics is an independent health economics research organisation. During my internship, I developed an automated data collection tool to support health economists.

- Cut data collection time from 2 days to 10 minutes by building an automated pipeline to collect pharmaceutical regulatory data from the National Institute for Health and Care Excellence and the European Medicines Agency
- Engineered a RAG pipeline with Chroma and LangChain to improve retrieval and summarisation accuracy on medical documents
- Conducted hands-on training sessions allowing non-technical colleagues to operate the workflow independently

Acquisition & Distribution Assistant Manager, Danal Entertainment, South Korea

Apr 2019 – Oct 2022

Danal Entertainment distributes films, TV series, and music in the South Korean market. I supported commercial decision-making and managed the distribution of 100+ films and TV series across platforms including Netflix, Naver, and TVING

- Developed statistical forecasting models to predict film profitability across a 10,000+ film catalogue; achieved a 45–65% recoupment of licence fees at or before release
- Negotiated licensing agreements with 50+ international film providers, leveraging forecast insights to evaluate commercial opportunities and secure an exclusive distribution partnership with Coupang Play
- Prepared monthly and annual sales-performance reporting, translating complex data into acquisition and pricing recommendations for senior management

EDUCATION

MSc Data Science (Distinction), City St George's, University of London, UK

Sep 2023 – Jan 2025

- Dissertation (72%): Developed an interpretable ensemble anomaly detection framework for industrial control systems using DBSCAN labelling, LSTM-GRU stacking and SHAP/LIME explainability
- Modules: Natural Language Processing (88.5%), Machine Learning (81%), Big Data (78%), Neural Computing (76%), Visual Analytics (75%)

BA Economics and English Language & Literature (2:1), Dongguk University, Seoul, South Korea

Mar 2015 – Feb 2018